

- 7 (a) most α -particles deviated through small angles B1
 (accept 'undeviated')
 few α -particles deviated through angles greater than 90° B1 [2]
- (b) (i) allow $10^{-9} \text{ m} \rightarrow 10^{-11} \text{ m}$ B1 [1]
- (ii) allow $10^{-13} \text{ m} \rightarrow 10^{-15} \text{ m}$ B1 [1]
 (if (i) and (ii) out of range but (ii) = $10^{-4}(\text{i})$, then allow 1 mark)
 (if no units or wrong units but (ii) = $10^{-4}(\text{i})$, then allow 1 mark)